

Multi-Turn Speech Dialogue: Supplementary Papers (2022-2026)

A supplementary paper list for Multi-Turn Speech Dialogue, searched through 13 August 2026. Papers already listed in `readme_restructured.docx` are excluded.

Survey

1. "A Survey of Recent Advances on Turn-taking Modeling in Spoken Dialogue Systems". *Galo Castillo-Lopez et al.* IWSDS 2025. [Paper](#)

Datasets

1. "DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech". *Keon Lee et al.* arXiv 2022. [Paper](#)
2. "DOROTHIE: Spoken Dialogue for Handling Unexpected Situations in Interactive Autonomous Driving Agents". *Ziqiao Ma et al.* EMNLP Findings 2022. [Paper](#)
3. "SpokenWOZ: A Large-Scale Speech-Text Benchmark for Spoken Task-Oriented Dialogue Agents". *Shuzheng Si et al.* arXiv 2023. [Paper](#)
4. "CALLS: Japanese Empathetic Dialogue Speech Corpus of Complaint Handling and Attentive Listening in Customer Center". *Yuki Saito et al.* Interspeech 2023. [Paper](#)
5. "J-CHAT: Japanese Large-scale Spoken Dialogue Corpus for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2024. [Paper](#)
6. "Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation". *Se Park et al.* ACL 2024. [Paper](#)
7. "Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models". *Sehun Lee et al.* NAACL 2025. [Paper](#)
8. "RealTalk-CN: A Realistic Chinese Speech-Text Dialogue Benchmark With Cross-Modal Interaction Analysis". *Enzhi Wang et al.* arXiv 2025. [Paper](#)
9. "Exploring Emotional Nuances in Spoken Dialogue: Dataset Construction and Prediction of Emotional Dialogue Breakdown". *Hyuga Nakaguro et al.* IWSDS 2026. [Paper](#)
10. "DuplexChat: Constructing Speaker-Separated Full-Duplex Dialogue Speech at Scale for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2026. [Paper](#)
11. "Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Knowledge-Grounded Information Seeking". *Songbo Hu et al.* ACL Findings 2026. [Paper](#)

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. "Using Transition Duration to Improve Turn-taking in Conversational Agents". *Charles Threlkeld et al.* SIGDIAL 2022. [Paper](#)
2. "When can I Speak? Predicting Initiation Points for Spoken Dialogue Agents". *Siyan Li et al.* SIGDIAL 2022. [Paper](#)
3. "How Much Does Prosody Help Turn-taking? Investigations using Voice Activity Projection Models". *Erik Ekstedt et al.* SIGDIAL 2022. [Paper](#)
4. "Device Directedness with Contextual Cues for Spoken Dialog Systems". *Dhanush Bekal et al.* arXiv 2022. [Paper](#)
5. "Triadic Multi-party Voice Activity Projection for Turn-taking in Spoken Dialogue Systems". *Mikey Elmers et al.* Interspeech 2025. [Paper](#)
6. "Synchronization and Turn-Taking in Full-Duplex Speech Dialogue Models". *Pablo Riera et al.* arXiv 2026. [Paper](#)
7. "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions". *Dongwook Lee et al.* ACL 2026. [Paper](#)
8. "Rethinking Binary Evaluation of Turn-Taking under Inherent Ambiguity". *Yunosuke Kubo et al.* SIGDIAL 2026. [Paper](#)

Evaluation Focus: Full-duplex Interaction

9. "FD-Bench: A Full-Duplex Benchmarking Pipeline Designed for Full Duplex Spoken Dialogue Systems". *Yizhou Peng et al.* Interspeech 2025. [Paper](#)
10. "tau-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains". *Soham Ray et al.* ICML 2026. [Paper](#)
11. "Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency". *Guan-Ting Lin et al.* arXiv 2026. [Paper](#)
12. "Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge". *Chengyou Wang et al.* arXiv 2026. [Paper](#)
13. "Full-Duplex-Bench-v2: A Multi-Turn Evaluation Framework for Duplex Dialogue Systems with an Automated Examiner". *Guan-Ting Lin et al.* ACL 2026. [Paper](#)
14. "MTR-DuplexBench: Towards a Comprehensive Evaluation of Multi-Round Conversations for Full-Duplex Speech Language Models". *Zhang He et al.* ACL Findings 2026. [Paper](#)
15. "M3-DuplexBench: A Multi-Turn, Multilingual, Multidomain Benchmark for Full-Duplex Spoken Dialogue Models". *Ryo Fukuda et al.* arXiv 2026. [Paper](#)

Evaluation Focus: Other Dialogue Capabilities (其他对话能力)

16. "Benchmarking Open-ended Audio Dialogue Understanding for Large Audio-Language Models". *Kuofeng Gao et al.* arXiv 2024. [Paper](#)
17. "SOVA-Bench: Benchmarking the Speech Conversation Ability for LLM-based Voice Assistant". *Yixuan Hou et al.* Interspeech 2025. [Paper](#)

18. "MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols". *Yuhao Du et al.* arXiv 2025. [Paper](#)
19. "VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?". *Dhruv Jain et al.* arXiv 2025. [Paper](#)
20. "C3: A Bilingual Benchmark for Spoken Dialogue Models Exploring Challenges in Complex Conversations". *Chengqian Ma et al.* EMNLP 2025. [Paper](#)
21. "The Context Trap: Why End-to-End Audio Language Models Fail Multi-turn Dialogues". *Zhi Rui Tam et al.* IWSDS 2026. [Paper](#)
22. "SDiaReward: Modeling and Benchmarking Spoken Dialogue Rewards with Modality and Colloquialness". *Jingyu Lu et al.* ACL 2026. [Paper](#)
23. "SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?". *Jonggeun Lee et al.* ACL 2026. [Paper](#)
24. "Style Amnesia: Investigating Speaking Style Degradation and Mitigation in Multi-Turn Spoken Language Models". *Yu-Xiang Lin et al.* ACL Findings 2026. [Paper](#)

Models

Interaction Mode: Turn-taking

1. "Generative Spoken Dialogue Language Modeling". *Tu Anh Nguyen et al.* TACL 2023. [Paper](#)
2. "LLaMA-Omni: Seamless Speech Interaction with Large Language Models". *Qingkai Fang et al.* arXiv 2024. [Paper](#)
3. "Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction". *Ailin Huang et al.* arXiv 2025. [Paper](#)
4. "Baichuan-Audio: A Unified Framework for End-to-End Speech Interaction". *Tianpeng Li et al.* arXiv 2025. [Paper](#)
5. "Kimi-Audio Technical Report". *KimiTeam et al.* arXiv 2025. [Paper](#)
6. "Qwen3-Omni Technical Report". *Jin Xu et al.* arXiv 2025. [Paper](#)
7. "PACHAT: Persona-Aware Speech Assistant for Multi-party Dialogue". *Dongjie Fu et al.* EMNLP 2025. [Paper](#)
8. "VoxMind: An End-to-End Agentic Spoken Dialogue System". *Tianle Liang et al.* ACL 2026. [Paper](#)
9. "ZipVoice-Dialog: Non-Autoregressive Spoken Dialogue Generation with Flow Matching". *Han Zhu et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Half-duplex / Barge-in

10. "Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities". *Zhifei Xie et al.* arXiv 2024. [Paper](#)
11. "VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large Speech-Language Model". *Zuwei Long et al.* arXiv 2025. [Paper](#)
12. "AV-Dialog: Spoken Dialogue Models with Audio-Visual Input". *Tuochao Chen et al.* arXiv 2025. [Paper](#)

13. "SoulX-Duplug: Plug-and-Play Streaming State Prediction Module for Realtime Full-Duplex Speech Conversation". *Ruiqi Yan et al.* arXiv 2026. [Paper](#)

Interaction Mode: Full-duplex

14. "Duplex Conversation: Towards Human-like Interaction in Spoken Dialogue Systems". *Ting-En Lin et al.* arXiv 2022. [Paper](#)
15. "A Full-duplex Speech Dialogue Scheme Based On Large Language Models". *Peng Wang et al.* NeurIPS 2024. [Paper](#)
16. "Towards a Japanese Full-duplex Spoken Dialogue System". *Atsumoto Ohashi et al.* Interspeech 2025. [Paper](#)
17. "FLM-Audio: Natural Monologues Improves Native Full-Duplex Chatbots via Dual Training". *Yiqun Yao et al.* arXiv 2025. [Paper](#)
18. "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations". *Junjie Chen et al.* arXiv 2025. [Paper](#)
19. "Incorporating Dialogue State Tracking into Japanese Full-duplex Task-oriented Spoken Dialogue Model". *Yuya Chiba et al.* IJCNLP-AACL 2025. [Paper](#)
20. "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction". *Junbo Cui et al.* arXiv 2026. [Paper](#)
21. "DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action". *Haoyang Zhang et al.* arXiv 2026. [Paper](#)
22. "BayLing-Duplex: Native Full-Duplex Speech Dialogue with a Single Autoregressive LLM". *Qingkai Fang et al.* arXiv 2026. [Paper](#)
23. "F-Actor: Controllable Conversational Behavior in Full-Duplex Models". *Maike Zuefle et al.* ACL Findings 2026. [Paper](#)
24. "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs". *Zhenyu Liu et al.* ACL 2026. [Paper](#)
25. "JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents". *Yinhao Bai et al.* arXiv 2026. [Paper](#)

Training Methods

Interaction Mode: Turn-taking

1. "Adapting Text-based Dialogue State Tracker for Spoken Dialogues". *Jaeseok Yoon et al.* DSTC 2023. [Paper](#)
2. "Approaching Dialogue State Tracking via Aligning Speech Encoders and LLMs". *Simon Sedlacek et al.* arXiv 2025. [Paper](#)
3. "The Speech-LLM Takes It All: A Truly Fully End-to-End Spoken Dialogue State Tracking Approach". *Nizar El Ghazal et al.* arXiv 2025. [Paper](#)
4. "WavAlign: Enhancing Intelligence and Expressiveness in Spoken Dialogue Models via Adaptive Hybrid Post-Training". *Yifu Chen et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Half-duplex / Streaming

5. "LLM-Enhanced Dialogue Management for Full-Duplex Spoken Dialogue Systems". *Hao Zhang et al.* arXiv 2025. [Paper](#)
6. "Dual-Reasoner: Bridging Interleaved Atomicity and Streaming Latency via Thinking-while-Talking". *Yangzhuo Li et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Full-duplex

7. "Reinforcement Learning Enhanced Full-Duplex Spoken Dialogue Language Models for Conversational Interactions". *Chen Chen et al.* COLM 2025. [Paper](#)
8. "Chronological Thinking in Full-Duplex Spoken Dialogue Language Models". *Donghang Wu et al.* arXiv 2025. [Paper](#)
9. "Reproducing Proficiency-Conditioned Dialogue Features with Full-duplex Spoken Dialogue Models". *Takao Obi et al.* IWSDS 2026. [Paper](#)
10. "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model". *Yuto Abe et al.* IWSDS 2026. [Paper](#)
11. "Dual-Axis Generative Reward Model Toward Semantic and Turn-taking Robustness in Interactive Spoken Dialogue Models". *Yifu Chen et al.* ACL 2026. [Paper](#)
12. "Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models". *Kyudan Jung et al.* ACL Industry Track 2026. [Paper](#)